How people stop or slow the release of engineered organisms

HOW PEOPLE STOP OR SLOW THE RELEASE OF ENGINEERED ORGANISMS

A Global How-To for Anyone Starting From Zero

BEFORE YOU START: WHAT THIS GUIDE IS

This guide is written for one person sitting at a table with a phone or a laptop, who has just found out that something engineered is going to be grown, released, imported, or fed to people near them, and who does not know what to do next.

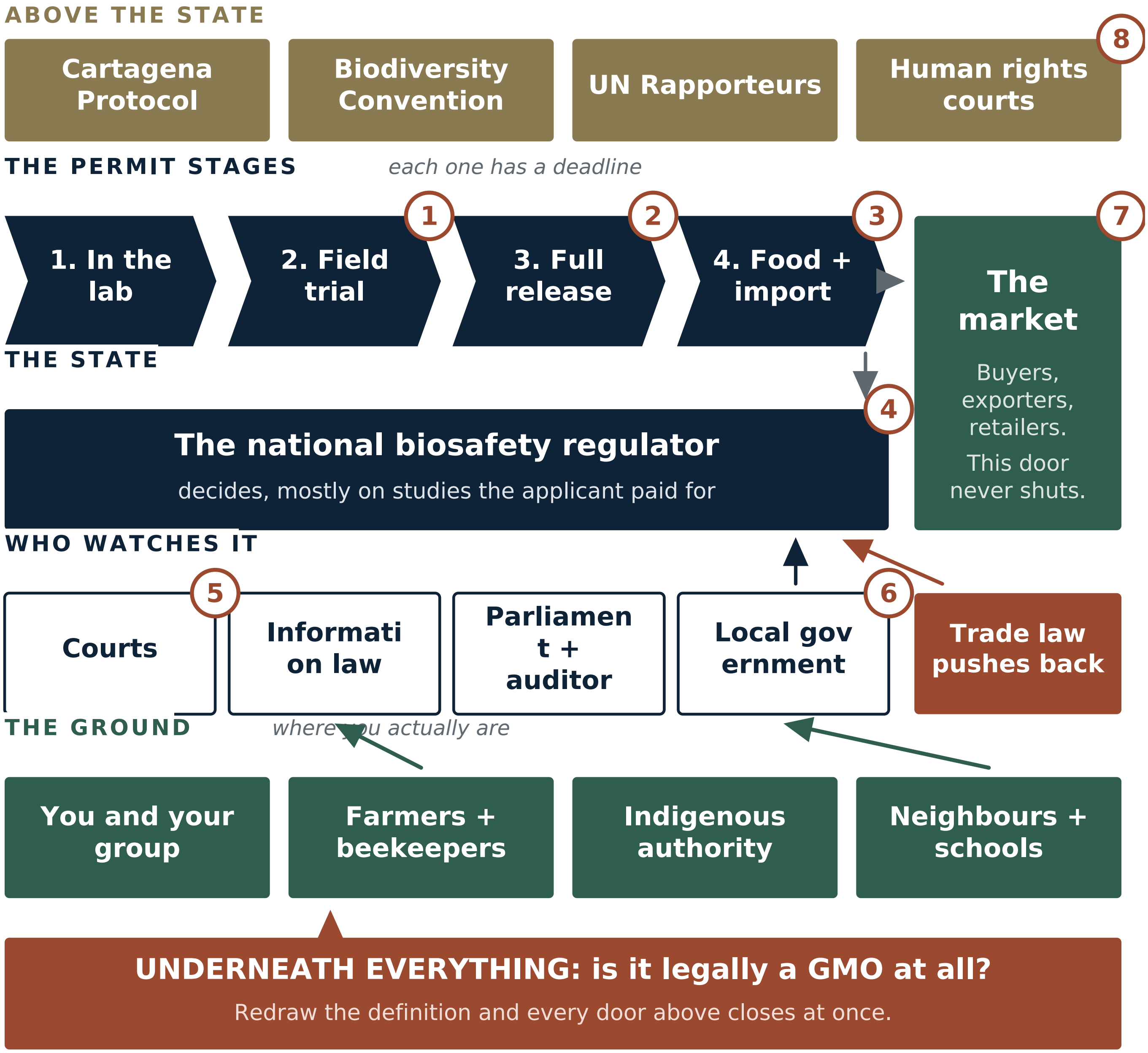
You do not need a science degree. You do not need money. You do not need to belong to an organisation. Most of the work in this guide is reading public paperwork, writing plain letters before a deadline, and getting other people to sign their names next to yours.

The guide is global. There is no world government for this, no single office that decides. Instead there are about a hundred national systems that all work in roughly the same shape, plus a handful of international treaties that sit above them. Once you learn the shape, you can find your own country's version of each piece in an afternoon. Every section below tells you what the piece is called in several countries so you can recognise yours.

Read the glossary at the end first if the words are new. It is one page.

THE WHOLE FIELD, ON ONE PAGE

Most people have never seen the machinery that decides these things, so it is hard to know where you would even push. Figure 1 is the whole playing field. Everything else in this guide is a close-up of one part of it, and every later diagram uses this same layout with the rest faded back, so you can always see where you are standing.



THE EIGHT DOORS - the places you can push

1. Comment before the deadline
2. Comment on the full release
3. Import approval: often the only door
4. Ask for the documents; appeal a refusal
5. Judicial review, on process
6. A council motion: cheap and public
7. Buyers: no rules, no deadline
8. International bodies: free and slow

Figure 1. The whole field, and the eight places you can push into it. Read it in four movements. **Left to right:** the four permit stages an organism passes through, and the market it ends up in. **Top down:** the international layer, which sits above your government, is free to write to, and cannot overrule it. **Bottom up:** the people a release actually touches, pushing into the bodies that watch the regulator. **And across the bottom:** the definition layer, which decides whether the organism enters this system at all.

Four things are worth noticing before you read any further.

The pipeline runs left to right, and it gets harder as it goes. Stopping something at the field-trial stage is a different job from stopping it after it is planted across a country. Everything you can do early is worth more than the same effort later.

Every permit stage has a deadline; the market has none. That is why the green column on the right never closes. It is also why a group with no money and no lawyer still has somewhere to push after a decision has gone against them.

The doors are numbered in order of cheapness, not importance. Doors 1 to 4 cost postage and evenings. Door 5 costs years. Most people skip doors 1 to 4 and then wish they had a lawyer.

The bar across the bottom is the newest problem. If your government decides that a gene-edited organism is not legally a GMO, it never enters the pipeline, no comment window ever opens, and doors 1 to 4 are simply not there. Watch the definitions as closely as you watch the permits.

WHAT THIS GUIDE WILL NOT PRETEND

A guide that flatters you is useless. Here is what is against you, and what arguments will lose.

The "it will poison you" argument will lose, and it will cost you. The large scientific and food-safety bodies that have reviewed approved GM foods — including national academies of science and the food-safety agencies of the countries that grow the most — have concluded that the GM foods currently approved and on the market are not more dangerous to eat than their conventional equivalents. You can disagree with how those reviews were done. But if you build a campaign on a health scare, the other side will spend one paragraph on it, win it, and then use it to make everything else you said look unserious. Judges notice. Reporters notice. Neighbours notice.

The arguments that survive contact with a regulator or a judge are these:

- **Consent.** The people who live, farm, fish, or hold rights where the release happens were not asked, or were asked in a way that does not count.
- **Contamination and coexistence.** Pollen, seed, escaped fish and spilled grain move. Once an engineered trait is out, it is very often not retrievable. Ask who has to keep whom out, and who pays when mixing happens.
- **Liability.** If a neighbouring organic farm loses its certification, or a shipment is rejected at a foreign port, who is legally responsible? In most countries the honest answer is nobody in particular.
- **Monitoring.** Almost every approval promises post-release monitoring. Almost no country funds it properly. Ask for the monitoring plan and the monitoring reports. The gap between them is often the strongest thing in your file.
- **Herbicides and resistance.** Most engineered crop area worldwide is herbicide-tolerant. What happens to herbicide volumes, to weed resistance, and to the people spraying is a documented, measurable question.
- **Ownership.** Patents on seed, contracts that forbid saving seed, and the concentration of the global seed and pesticide market in a handful of firms — Bayer, Corteva, Syngenta Group and BASF chief among them — are matters of public record and public interest.
- **Irreversibility.** For anything designed to spread on its own — gene drives above all — "we can recall it" is not available. That changes the standard of care that should apply.

The scale is against you. Engineered crops cover well over 200 million hectares across roughly thirty countries and the area is still growing. You are not going to reverse that. You may be able to stop one release, in one place, this year.

The realistic outcome is delay, conditions, and a record. Delay is not nothing. It costs money, it moves elections, it lets evidence accumulate, and it sometimes outlasts the company's interest in the product. Several engineered crops died not from a ban but because buyers said no while the approvals were still pending.

The rules can be changed underneath you. This is the newest problem and the biggest one. Instead of approving engineered organisms one by one, governments are redefining which ones count as engineered at all. Argentina, Brazil, Japan, England and others have moved gene-edited organisms outside the old GMO rules. The European Union adopted Regulation (EU) 2026/1388 on new genomic techniques, which entered into force on 16 July 2026 and applies from 17 July 2028, and which splits gene-edited plants into two categories — a lighter-touch "NGT-1" tier and an "NGT-2" tier that keeps closer to the old rules. In the United States, the 2020 SECURE rule was vacated by a federal court on 2 December 2024, the department reverted to the older regulations, and a replacement framework was still being consulted on in mid-2026.

If you only learn to fight permits, you will keep fighting permits while the category quietly empties out. Watch the definitions.

HOW THE SYSTEM WORKS

You cannot pull a lever you cannot see. Before anything else, work out which of the following stages your particular release is at, because each stage has its own door, its own paperwork, and its own deadline.

The five stages every release passes through

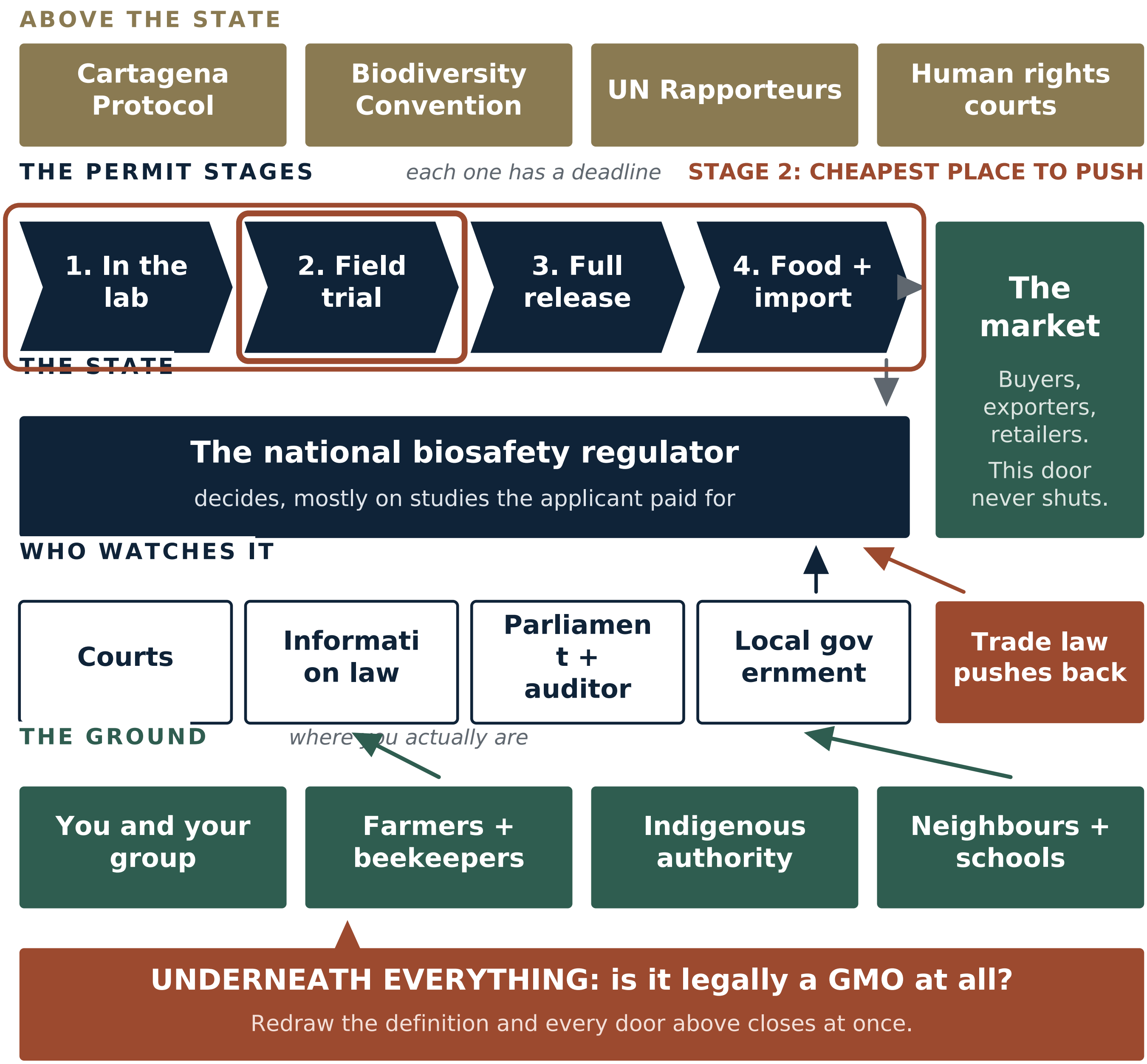


Figure 2. Where you are in the pipeline. The stage decides which door is open, how much the applicant has already spent, and how much of the organism is already outside. Stage 2 is the cheapest place in the whole field to stop something: the trial is small, the permit is specific, the location is often published, and very little money has been committed yet.

Stage 1 — Contained use. Work inside a lab or greenhouse. Usually approved by an institutional committee at the university or company, sometimes registered with a national authority. The records are thin but they exist, and university committee minutes are often obtainable.

Stage 2 — Field trial / confined release. The organism goes outdoors in a limited, supposedly controlled way. **This is the single best place to intervene.** The trial is small, the permit is specific, the location is often published, and the applicant has not yet spent much money. Many countries run a public comment period at this stage.

Stage 3 — Environmental release / cultivation approval. Permission to grow or release commercially, without the limits.

This is the big one, and it almost always has a formal public consultation attached.

Stage 4 — Food, feed and import approval. Separate from cultivation. A country that grows nothing engineered may still import a great deal of it. Import approvals have their own comment windows and are frequently the only door available in countries that do not grow the crop.

Stage 5 — The market. Retailers, food manufacturers, exporters, grain buyers and restaurant chains all make their own decisions and answer to customers, not regulators. This stage has no deadlines and no legal requirements, which makes it the one door that is never closed.

Underneath all five sits a sixth question: **is this thing legally a "GMO" at all?** If a gene-edited organism has been reclassified as conventional in your country, stages 2 to 4 may not apply to it. Find out early, because it decides which of the tools below you have.

Who decides, and who watches them

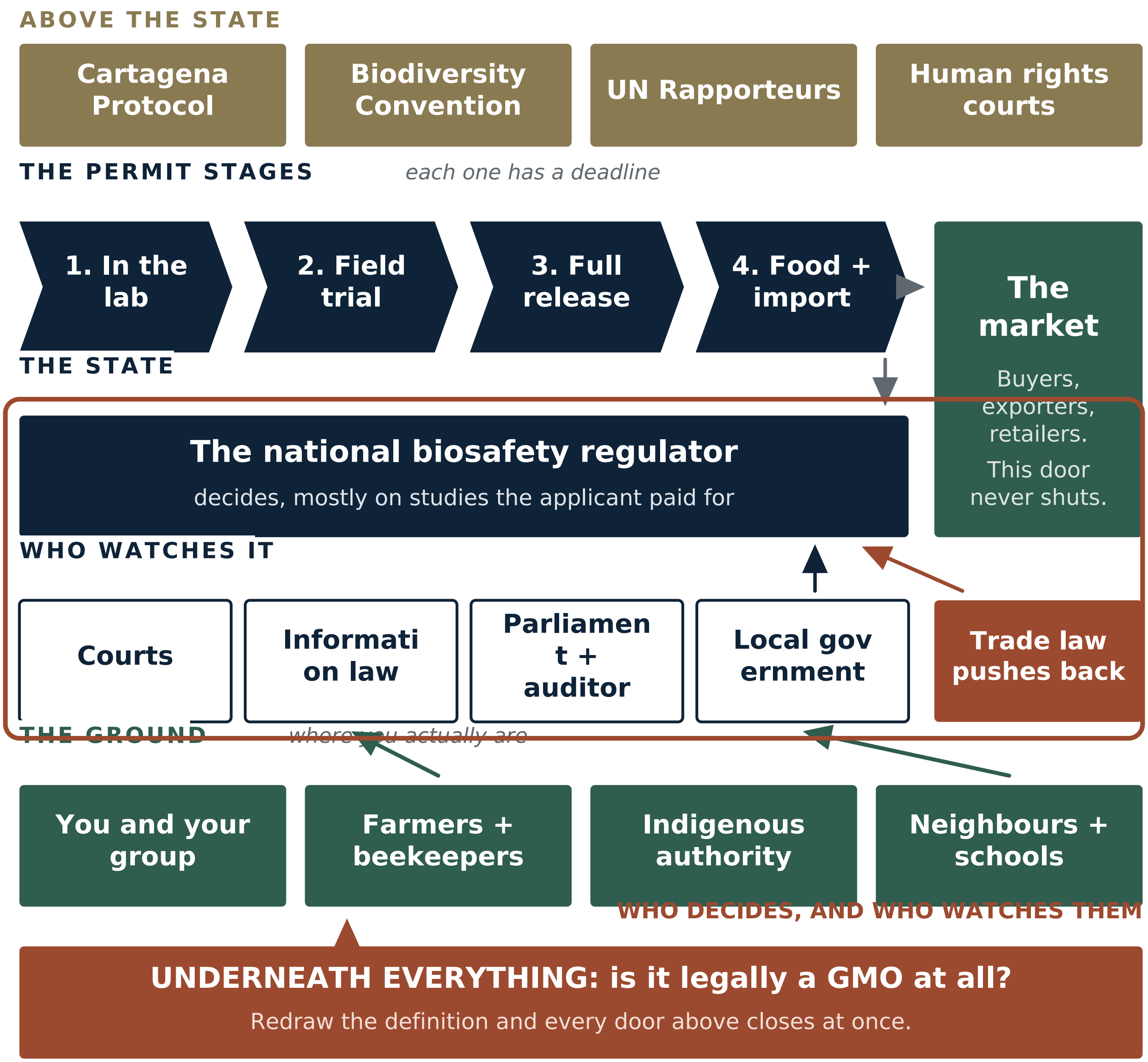


Figure 3. The state, the four bodies that watch it, and what leans on it from outside. The regulator decides, but it mostly reviews studies the applicant paid for. None of the four overseers works for it. Trade law pushes back when a government restricts too much, and the international layer above can embarrass a state but cannot overrule it.

Find your country's version of the regulator

The names differ from country to country; the job is the same.

- **United States** — three agencies split it: the Department of Agriculture's Animal and Plant Health Inspection Service (plants and permits), the Environmental Protection Agency (anything engineered to act as a pesticide), and the Food and Drug Administration (food and animals). Notices and comment periods appear in the Federal Register; you comment at regulations.gov.
- **European Union** — the European Food Safety Authority assesses, the European Commission and member states decide. EFSA runs a public comment period, usually 30 days, after it publishes its opinion on an application. Its applications and documents are reachable through Connect.EFSA and OpenEFSA.
- **Australia** — the Office of the Gene Technology Regulator. It publishes a Risk Assessment and Risk Management Plan for each licence and opens it for comment for at least 30 days, advertised in the Government Notices Gazette, with a free email alert list. Its register also publishes the locations of active field trial sites — one of the few in the world that does.
- **India** — the Genetic Engineering Appraisal Committee, under the environment ministry. Its meeting minutes are published.
- **Philippines** — the Department of Agriculture and the Bureau of Plant Industry, with public notices at the local level for field trials.
- **Brazil** — CTNBio. **Canada** — the Canadian Food Inspection Agency and Health Canada. **Japan** — the agriculture and environment ministries. **New Zealand** — the Environmental Protection Authority, which runs full public hearings.
- **Everywhere else** — look for the "National Biosafety Authority", "Biosafety Committee", or the biosafety unit of the environment ministry. The fastest way to find it is described in the next section.

The layer above the countries

The Cartagena Protocol on Biosafety. An international agreement under the Convention on Biological Diversity, adopted in 2000, in force since 2003, with 173 parties. The United States is not one of them. It matters to you for three reasons:

- It creates the **advance informed agreement** procedure: before a living modified organism is exported to be released into another country's environment for the first time, the exporter must notify and the importing country must decide. That decision is a document, and documents can be demanded and challenged.
- It creates the **Biosafety Clearing-House** at bch.cbd.int — a free, public, searchable database of national decisions, risk assessments, laws and contact details for every party's competent national authority. If you do not know who decides in your country, this is where you find out, in about five minutes.
- Its **Article 26** allows countries to take socio-economic considerations into account — the effects on indigenous and local communities in particular. Most countries ignore this. It is available to be asked about.

The Nagoya–Kuala Lumpur Supplementary Protocol deals with liability and redress if a living modified organism causes damage. Fewer countries have ratified it. Ask whether yours has, and what its national liability rules actually say. The answer is usually embarrassing.

The Convention on Biological Diversity itself handles synthetic biology and gene drives at its Conferences of the Parties. Decision 16/21, adopted on 1 November 2024, set up an expert group on synthetic biology and pointed towards a thematic action plan for consideration at the next Conference of the Parties. These processes accept submissions from civil society. They are slow and they are not binding, but they are a place where a small group's document can end up in front of nearly every government at once.

The International Treaty on Plant Genetic Resources for Food and Agriculture recognises farmers' rights, including the right to save and exchange seed. Useful when the argument is about seed ownership rather than biology.

United Nations Special Rapporteurs — on the right to food, on toxics and human rights, and on the rights of indigenous peoples — accept complaints from anyone, free, by email. They cannot order anything. They can write to a government and publish the letter, which several governments dislike a great deal.

For indigenous peoples specifically: free, prior and informed consent is a standard recognised in the UN Declaration on the Rights of Indigenous Peoples and enforced in binding judgments by regional human rights courts. A release on or affecting indigenous territory without that consent is a much stronger case than the same release elsewhere. If this is your situation, this is your strongest single lever and it should be the centre of everything you file.

The layer that fights back

Trade law. The World Trade Organization's agreements on sanitary and phytosanitary measures and on technical barriers to trade, plus regional agreements, all restrict how far a country may go in restricting an approved product without a scientific justification the trade panel accepts.

Mexico is the clearest recent example, and worth knowing in full because it shows both the counterattack and the answer to it. Farmers, scientists and beekeepers filed a collective lawsuit against genetically modified maize in 2013, which froze planting permits for years. A presidential decree later restricted GM corn for tortillas. The United States challenged the restrictions under the trade agreement between the United States, Mexico and Canada, and on 20 December 2024 the dispute panel ruled against Mexico. Mexico did not simply comply: on 17 March 2025 it published an amendment to Articles 4 and 27 of its own constitution prohibiting the planting of genetically modified maize.

Know this in advance so it does not surprise you, and so you frame your demands in the forms that survive it: process requirements, consent requirements, labelling and traceability, liability rules, and the protection of centres of crop origin and of indigenous rights, rather than blanket import bans.

Follow the permit, the window, and the money

Behind every release there is an applicant (a company, a university, a public institute, sometimes a foreign aid programme), a regulator who decides, and a funder who pays. Three questions unlock most cases:

1. **What does the application not answer?**
2. **When does the window to say so close?**
3. **Who else loses if this goes ahead, and would they sign a letter?**

QUICK REFERENCE: WHAT EACH STEP IS WORTH

Be honest with yourself before you spend a year on this. The table below is a judgement drawn from how documented campaigns have gone. **It is not measured data, and no one has run a controlled study of this.** Treat the numbers as a rough ranking of effort against effect, not as a prediction about your case.

"Success" here means the release is stopped, delayed, conditioned, or forced back for more assessment — not that the technology goes away.

What you do	Rough odds it changes something	Time	Cost
Read the file and do nothing else	near zero	days	none
File one plain public comment	5–10%	hours	none
File a detailed technical comment with sources	15–30%	weeks	none
Same, signed by 20+ affected people and 2 scientists	25–45%	1–3 months	near none
Freedom of information request for the missing data	20–40%	1–6 months	none to small
Get one credible news story	20–35%	weeks	none
Local council motion or GMO-free zone declaration	25–50% locally	months	none to small
Buyer / exporter / retailer pressure	20–45%	months	none
Court challenge on process grounds	20–40%	1–4 years	high without a free lawyer
International body submission (treaty, Rapporteur)	10–25%	months to years	none
All of the above, run together	**40–65%**	**1–5 years**	**low to moderate**
Holding the win through appeal and re-application	**20–45%**	**many years**	**low to moderate**

Two things stand out in that table and both are worth saying plainly.

The cheapest step is the most skipped. Almost every regulator in the world runs a public comment period, and in most cases it receives a handful of submissions, most of them one-line objections that the agency can dismiss in a single sentence. A twelve-page comment with page references, a named gap in the risk assessment, and twenty signatures from people who farm within five kilometres of the site is a different object entirely. It has to be answered on the record. That answer is then something a court can read.

The finish line is not where you think. Look at the last row. Winning a decision starts an appeal, a re-application with slightly different paperwork, or a change in the rules that makes your win irrelevant. Plan for the second lap from the first day, and do not disband the group when you win.

THE FIVE STEPS, AND WHY THEY RUN TOGETHER

The five steps are not a queue.

Every line between them is a real dependency.

Figure 4. The five steps are numbered for reading, not for doing. Every line in this diagram is a real dependency: the file makes the comment credible, the comment makes the story printable, the story makes the council motion pass, and the motion makes the agency answer on the record.

The next five sections take them one at a time. Start all of them in the first fortnight and let each one feed the others.

STEP 1: FIND THE EXACT THING YOU ARE OPPOSING

Vague opposition goes nowhere. “I am against GMOs” is a mood. “Application number 24-117-01r, for a field trial of herbicide-tolerant engineered sugarcane on 40 hectares at [place], open for public comment until 14 March, whose risk assessment contains no gene flow data for the wild relative growing along the river 300 metres away” is a case.

Getting from the first to the second takes about a week of evenings.

The five questions

1. What exactly is being released, and where?

Get the species, the trait, the event name (engineered organisms get codes like MON 810 or DP-3Ø4423-9), the hectares or the number of animals, and the location. Location matters more than anything else on this list.

2. Which permit is it, and which window is open right now?

Every application has a file number and almost every one has a deadline. Write the deadline on the wall. Missing it costs you the cheapest opportunity you will ever get. If the window is closed, ask in writing whether it can be reopened, and start on the stages that have no deadlines: information requests, the market, the press, and the local council.

3. What does the application not answer?

This is the heart of the work and Step 2 covers it in detail. You are not looking for a smoking gun. You are looking for a question the applicant did not ask, a study they did not do, a plan they promised and did not attach, or an organism in the neighbourhood they did not mention.

4. Who else is affected, and would they put their name on it?

Organic and certified farmers within the pollen or seed range. Beekeepers. Seed savers. Neighbouring landholders. Exporters who sell into markets that refuse the trait. Indigenous peoples and their own authorities. Fishers, if the release is aquatic. Local water users. A comment signed by these people carries weight a comment from a stranger does not.

5. Who decides, who profits, and who is watching?

Name the deciding official or committee. Name the applicant and its parent company. Find out who funds the programme — public research money and foreign aid budgets both come with their own accountability channels. And list who already watches this: the environment committee of your parliament, the national auditor, the ombudsman, the national human rights institution, the farming press, the export trade bodies.

Worked example

Say you hear that engineered mosquitoes will be released near your town.

- **What:** find the species, the modification, the number, the release site, and whether it is self-limiting or self-spreading. That last distinction decides how strong your case is. Self-spreading releases — gene drives — are designed not to be recoverable.
- **Which permit:** search your national biosafety authority's register and the Biosafety Clearing-House for the decision or the pending application. If the work is funded from abroad, search the funder's own grant database; it is often more informative than the regulator's file.
- **What is missing:** is there a monitoring plan and who pays for it? Is there a plan for stopping if something goes wrong, and is it technically possible? Were nearby communities consulted, when, and what were they told? Was any socio-economic assessment done, as Cartagena Article 26 permits?
- **Who else:** the villages within the flight range, the health workers, the local council, the beekeepers, the national indigenous organisation if any affected land is theirs.
- **Who decides:** the biosafety authority, the health ministry, the local government that hosts the site, and — a lever people forget — the ethics committee of the university running the trial and the funder's own safeguards office.

This is not hypothetical. Target Malaria, a research consortium developing genetically modified mosquitoes with work in several West African countries, held a trial at Souroukoudingan in Burkina Faso on 11 August 2025. Burkina Faso's authorities sealed the programme's facilities on 18 August 2025 and suspended all of its activities on 22 August 2025. A programme funded from three continents was stopped by objection raised where the release was actually happening.

Within a fortnight you have a target list instead of a worry. That is the difference this step makes.

STEP 2: BUILD THE FILE

Regulators, judges, reporters and buyers all move on documents. Build yours in three layers. Save everything as a dated PDF and keep a copy somewhere that is not your laptop.

Layer 1 — the paperwork the decision rests on

Get, and read:

- **The application itself**, including its annexes. This is usually public, sometimes only on request, and often partly blacked out.
- **The risk assessment.** Note who wrote it. In most systems the applicant supplies the studies and the agency reviews them; independent studies are the exception, not the rule. That is a legitimate thing to point out.
- **The monitoring plan**, and separately, **the monitoring reports** from any previous approval of the same trait in the same country. The gap between the plan and the reports is where a lot of cases are won.
- **The consultation record.** Who was notified, how, in what language, with how much notice.
- **Any redactions.** "Confidential business information" claims are challengeable. Ask what legal basis was used for each redaction. Sometimes asking is enough to get some of it released.

Layer 2 — the ground

The applicant's file describes the site in the abstract. Your job is to describe it in the concrete, because you are there and they are not.

- **Map what is actually around the site.** Other farms and what they grow. Certified organic operations and their certification requirements. Apiaries. Seed banks and seed-saving households. Schools, houses and water sources. Protected areas.
- **Find the wild and weedy relatives.** For nearly every engineered crop there is a related plant it can cross with. Where that relative grows near the site is a specific, checkable fact, and its absence from the risk assessment is a specific, checkable omission.
- **Establish whether you are in a centre of origin or diversity** for the crop — the regions where it was first domesticated and where its landrace varieties still exist. Maize in Mexico and Central America, potato in the Andes, rice in Asia, sorghum and coffee in Africa, wheat in the Fertile Crescent. Releases in a centre of origin are treated more seriously by every international body, and rightly.
- **Photograph and date everything.** Photos with dates and locations are evidence. Screenshots of web pages should be archived at a service such as the Internet Archive so the page cannot quietly change later.
- **Record testimony.** Written statements from farmers, elders, beekeepers and local officials, signed and dated, are usable in almost every forum.

Layer 3 — the gap

Now write down what is missing. Each of these is a separate, usable ground:

- No independent data — every study cited was produced or funded by the applicant.
- No gene flow assessment for a relative that exists nearby.
- No coexistence rules, so no answer to how a neighbouring farm stays clean.
- No liability regime, so no answer to who pays if it does not.
- No monitoring plan, or a monitoring plan with no funding, no method, and no end date.
- No consultation with affected communities, or consultation held in the wrong language, at the wrong time, after the decision was effectively made.
- No socio-economic assessment where the law allows one.
- No consideration of what happens to herbicide use, or to resistance in the target pest.
- For anything self-spreading: no plan to stop or reverse it, because none exists.

A file that says "we think this is dangerous" gets one paragraph in reply. A file that says "the risk assessment at page 41 states there are no sexually compatible relatives within the release area; the attached photographs, dated and located, show that relative growing 300 metres from the site boundary" has to be dealt with.

Organise it so it persuades

Keep three things from day one:

- **A master timeline.** Every application date, every notice, every letter you sent, every reply. This becomes the spine of any legal filing and of any reporter's story.

- **A document index.** One line per document: what it is, where it came from, what date, and what it proves. So any claim can be sourced in seconds.
- **An argument bank.** Covered below.

WHERE TO ACTUALLY FIND THE DOCUMENTS

This is the practical companion to Step 2. Start at the top of the list and stop when you have what you need.

The Biosafety Clearing-House — bch.cbd.int. Free, global, and the single best starting point. It holds national decisions on living modified organisms, risk assessments, national biosafety laws, and the name and email address of the competent national authority in every country that is a party to the Cartagena Protocol. If you do not know who to write to, you will know in five minutes.

Your national register. Names vary:

- United States: the Federal Register (notices and comment periods) and regulations.gov (where you file). The agriculture department's permit and notification database lists field trial permits.
- European Union: the EU register of authorised GM food and feed, the EFSA journal, Connect.EFSA for applications and OpenEFSA for documents. EFSA opens a public comment period of about 30 days once its opinion is published — put a reminder in your calendar the moment an application appears.
- Australia: the OGTR GMO Record, which lists every licence and the locations of active field trial sites, with a free email alert list and notices in the Government Notices Gazette.
- India: GEAC meeting agendas and minutes on the environment ministry site.
- Canada, Brazil, Japan, New Zealand, South Africa, Kenya, Nigeria, Bangladesh, the Philippines: each publishes approvals somewhere on the biosafety authority's website, and all of them also file to the Biosafety Clearing-House, which is often easier to search.

Freedom of information. Most countries have a law that lets any person request government records, often including non-citizens. It is usually free or close to it, and it usually has a legal deadline for the reply. Ask narrowly and specifically — a request for "all documents about GMOs" gets refused; a request for "the monitoring reports submitted under condition 7 of licence DIR-192, for the years 2023 to 2025" gets answered.

The company's own words. Investor presentations, annual reports and stock exchange filings say things in public that the permit application does not. Patents — searchable free at Espacenet, Google Patents and WIPO Patentscope — describe exactly what an organism is engineered to do, often years before a permit application appears.

The funder. Public research councils, development agencies and foundations publish grant databases with project descriptions, budgets and timelines. Many have their own complaints and safeguards procedures, entirely separate from the biosafety regulator, and far less used.

The institution. Universities have institutional biosafety committees and research ethics committees whose minutes are often obtainable, especially at public universities subject to freedom of information law.

Contamination records. The GM Contamination Register, compiled by GeneWatch UK and Greenpeace, logged 396 incidents of contamination, illegal planting and negative agricultural side effects across 63 countries between 1997 and 2013 — roughly 25 a year. Named historical incidents are useful because they are documented and settled: StarLink maize entering the US food supply in 2000, and the LibertyLink rice contamination of the US long-grain rice crop in 2006, which led to settlements reported at around 750 million dollars and to export markets closing to American rice.

Existing groups. Somebody has almost certainly worked on your crop, your company, or your country before. Third World Network, GeneWatch UK, the Environmental Law Alliance Worldwide, national organic farming associations, and the farmer networks in your own region will often hand you their entire file for the cost of an email.

STEP 3: BUILD THE GROUP

Documents do not stop releases. People holding documents do.

Phase 1 — a core of three to five

You do not need a mass movement to begin. You need a small number of people who will still be there in eight months. Split the work explicitly, because undefined work does not get done:

- **Documents.** Reads the applications, tracks the deadlines, keeps the index.
- **Ground truth.** Visits the site, takes the photographs, talks to neighbours.
- **Writing.** Turns the file into comments, letters and submissions.
- **Outside contact.** Handles press, allies, lawyers and officials.
- **Keeping it going.** Calls the meetings, chases the tasks, notices when someone is burning out.

Agree at the start on what you want and what you will accept. “Stop it entirely” and “get a proper independent assessment and a liability rule” are different campaigns with different allies. Decide which one you are.

Phase 2 — widen, deliberately

Bring in the people whose names change how your submission is read:

- **Farmers**, especially certified organic ones, seed savers, and anyone exporting to a market that rejects the trait. Their loss is concrete, financial and documented, which is exactly what regulators respond to.
- **Beekeepers.** In several countries beekeepers have been the most effective objectors, because contaminated honey is a straightforward, provable commercial harm.
- **Indigenous peoples and their own authorities**, where affected — and through their own structures, never around them. If free, prior and informed consent applies, this becomes the centre of the case.
- **Local government.** A council motion costs a councillor very little and gives a national regulator something it must acknowledge.
- **Scientists.** You need two, not twenty. A named plant geneticist, ecologist, entomologist or agronomist willing to sign a two-page technical comment transforms the weight of everything you file. Ask retired academics; they have the expertise and none of the funding pressure.
- **A lawyer.** Environmental law clinics at universities, public interest law organisations, and networks such as the Environmental Law Alliance Worldwide take these cases without charge. Ask early — before a deadline passes, not after.
- **Buyers and exporters.** The trade body for your country’s grain, honey, seed or organic exports has a direct commercial interest and direct access to ministers. They are an unusual ally and often a decisive one.

Phase 3 — survive the years

Campaigns end from exhaustion far more often than from defeat.

- Meet on a fixed rhythm, even when nothing is happening.
- Give people finishable tasks. “Research the company” is not a task. “Download and summarise the 2024 annual report by Friday” is.
- Mark the small wins out loud: the comment filed, the information request answered, the deadline extended, the council motion passed.
- Rotate the boring jobs.
- Expect to be divided. Offers of money, jobs, trial payments to some farmers and not others, and side deals are the ordinary way these disputes are managed. Decide in advance how you will handle an offer, before one arrives.

Phase 4 — showing strength, and one hard rule

Turn people out for hearings, comment deadlines, council meetings and site visits. Numbers that can be counted are numbers that get reported.

And keep everything lawful. Do not destroy crops. Do not enter sites you have no right to enter. Do not damage equipment. This is not only about the law. Every documented case of crop destruction has been used, effectively, to reframe the dispute as one about vandalism rather than about the gap in the risk assessment — which is the one argument you were winning. It shifts sympathy to the applicant, it hands the press a different story, it can make your organisation liable for damages that will end it, and it discredits the farmers standing next to you. The strongest thing you have is that your file is accurate and your conduct is boring. Protect both.

STEP 4: THE LEGAL AND ADMINISTRATIVE ROUTES

In order of cost, cheapest first.

The public comment — the most underused tool in the world

Nearly every biosafety system on earth runs a public consultation, and in most of them the response is thin. A good comment does five things:

1. **Identifies itself precisely.** The application number, the applicant, the decision being consulted on.
2. **States who you are and why you are affected.** "I farm 12 hectares of certified organic maize 1.8 kilometres from the proposed site" outweighs several hundred words of general concern.
3. **Names a specific gap, with a page reference.** Not "this is unsafe" but "the risk assessment does not address X, at page N."
4. **Attaches evidence.** Photographs, maps, certification documents, a scientist's signed note.
5. **Asks for something specific and achievable.** An extension of the comment period. A public hearing. An independent assessment. A condition on the licence. A monitoring requirement with named funding. Refusal, if that is what you want — but always with a fallback ask, because an agency that will not refuse may well impose a condition.

Do not send a form letter. Agencies count identical submissions as one. Fifty different letters from fifty people beat five hundred copies of the same one.

Freedom of information

Cheap, slow, and often the thing that makes everything else possible. Request narrowly. If refused, appeal — most systems have an internal appeal and then an information commissioner or ombudsman, and refusals are overturned more often than people expect. A refusal is itself useful publicly: "the government will not release the monitoring reports" is a story.

Administrative appeal and reconsideration

Before court, most systems allow you to ask the agency to reconsider, or to appeal to a tribunal. It is far cheaper than litigation and it exhausts the domestic remedies that international bodies will later ask about.

Court

Understand what courts actually rule on. They very rarely decide whether an organism is safe. They decide whether the agency followed its own rules. The winning grounds, over and over, are:

- The agency skipped a step its own law required.
- The agency did not consult when it was supposed to.
- The agency's decision was not supported by the evidence in front of it.
- The agency did not apply the precautionary principle where its own constitution or statute required it.
- The decision-makers had conflicts of interest.

Recent examples worth citing in your own filings:

- **Philippines, 2024.** A farmer-led network, MASIPAG, together with Greenpeace Southeast Asia and others, petitioned the Supreme Court in October 2022 using the **writ of kalikasan** — a Philippine remedy protecting the constitutional right to a balanced and healthful ecology. The Court granted the writ in April 2023 and referred the case down for evidence. On 17 April 2024 the Court of Appeals revoked the biosafety permits for commercial propagation of Golden Rice and Bt eggplant. It did not rule that the crops were proven dangerous. It ruled that the government had not established the monitoring it was supposed to establish, that the precautionary principle applied where full scientific certainty was absent, and that the burden of evidence lay on the side seeking to change the status quo — which meant the agencies and institutes promoting the crops, not the farmers. The decision was reaffirmed in August 2024, and an appeal to the Supreme Court filed in November 2024 was still unresolved as this guide was written. Note what that means: the farmers won, twice, and are still in the fight.
- **India, 23 July 2024.** In Gene Campaign v. Union of India the Supreme Court split on whether to quash the approval of GM mustard, but agreed on two things: that decisions of the Genetic Engineering Appraisal Committee are open to judicial review, and that the government must develop a national policy on GM crops in consultation with states, farmers' groups and other stakeholders. It also addressed the verification of experts' credentials and conflicts of interest. That is a usable precedent well beyond India.

- **United States, 2 December 2024.** A federal court vacated the 2020 SECURE rule, which had exempted many engineered plants from review. The department reverted to its earlier regulations. A rule that deregulates can be challenged on how it was made.
- **India’s Bt brinjal moratorium, February 2010,** followed a series of public consultations held across the country after sustained pressure. It has never been lifted.

Court is expensive and slow. Do not start there. But know that a properly kept file from Step 2 is exactly what a lawyer needs to tell you whether you have a case, and that assessment is usually free.

Local and subnational routes

Often the fastest wins available. Local councils, provinces, states and regions have declared themselves GMO-free, restricted cultivation on public land, required notification of neighbours, or refused planning permission for facilities. Whether such a declaration is legally binding varies enormously — in some places it is enforceable, in others purely symbolic. Find out which yours would be before you spend six months on it. Even the symbolic ones generate press, put councillors on record, and create a map of refusal that national politicians read.

The international routes

Slow, non-binding, and worth doing anyway because they are free and because they create a record no domestic agency controls:

- **Cartagena Protocol.** Ask your national focal point, in writing, whether the advance informed agreement procedure was followed for this import, and whether socio-economic considerations under Article 26 were assessed. Copy the Biosafety Clearing-House.
- **UN Special Rapporteurs** on the right to food, on toxics, and on the rights of indigenous peoples. A short factual submission by email. They can write to your government and publish it.
- **The Convention on Biological Diversity’s synthetic biology process**, which accepts submissions ahead of its meetings — particularly relevant for gene drives and engineered organisms intended to spread.
- **Regional human rights systems** where indigenous or community land rights are involved. The Inter-American, African and European human rights bodies have all found states in violation for authorising activities on community land without proper consent.
- **The applicant’s own home country.** A company headquartered elsewhere may be subject to reporting obligations, national contact points under the OECD guidelines for multinational enterprises, or shareholder pressure that does not exist where the release is happening.

The market — no rules, no deadlines, no permission needed

This is where several engineered products actually died, and it is completely open to you.

Monsanto shelved its herbicide-tolerant wheat in 2004 after wheat buyers and growers’ organisations made clear they would not accept it, while the approval process was still running. The NewLeaf engineered potato was withdrawn after major fast food buyers declined to use it. Engineered salmon spent years struggling for shelf space because retailer after retailer publicly committed not to stock it.

Write to the buyers. Grain traders, food manufacturers, supermarket chains, restaurant groups, seed companies, and the export bodies that need access to markets that reject the trait. Their question is not “is it safe” but “will this cost me customers or shipments.” Answer that question for them, in one page, with the export data attached.

TURNING YOUR EVIDENCE INTO ARGUMENTS

The same fact means different things to different people. Learning to translate is what lets one small group reach every lever at once.

Take one documented fact: **the risk assessment claims there are no compatible wild relatives near the site, and you have dated photographs showing one 300 metres away.**

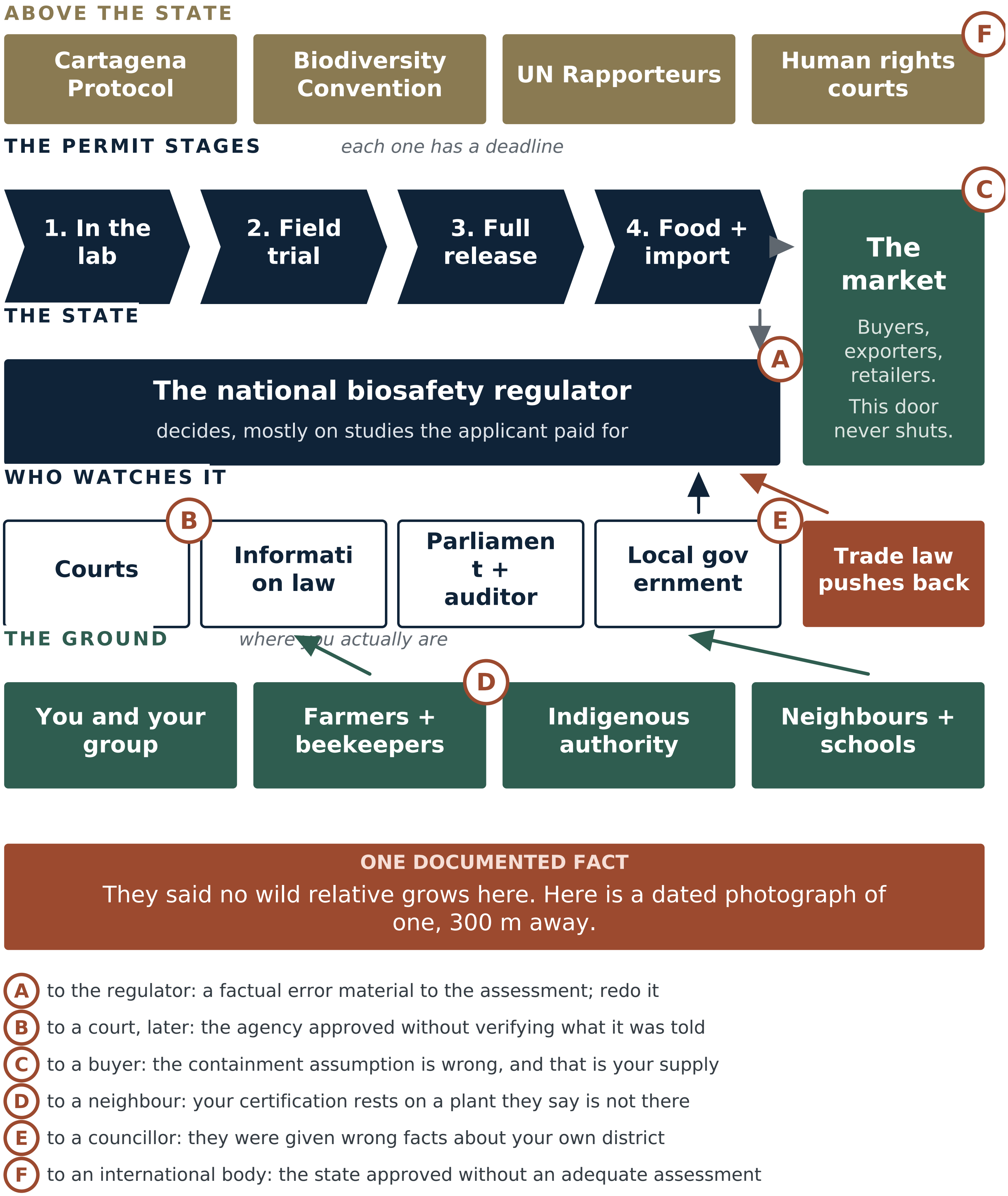


Figure 5. One documented fact, aimed at six positions on the same field. Nothing about the fact changes. What changes is which obligation it breaches, and whose problem that becomes.

- **To the regulator:** the application contains a factual error material to the gene flow assessment; the assessment must be redone before a decision.
- **To a court, later:** the agency approved a release on the basis of evidence it had reason to know was incomplete, and did

not verify it.

- **To a journalist:** they said it wasn't there. Here is a photograph of it, taken on this date, at this place.
- **To a neighbouring farmer:** your certification depends on this not happening, and the people who assessed the risk did not know the plant was there.
- **To an export buyer:** the containment assumption behind this approval has a documented error, which is a traceability risk for anything sourced from this region.
- **To a local councillor:** the national agency was given wrong information about your district and nobody local was asked.
- **To an international body:** the state approved a release without an adequate risk assessment, contrary to the standard its own law and the Cartagena Protocol set.

Same fact, seven arguments. Go through your five strongest facts and write out what each proves to each audience. That table is your argument bank. Keep every version anchored to the same documented facts, so that anything you say to a reporter can be proved on the record — the moment one claim cannot be sourced, everything else you have said is treated as equally unreliable.

Match the argument to the audience

- **Regulators** want a specific defect in a specific document, and a specific remedy. They do not want your view of the technology.
- **Courts** want procedure: what the law required, and what the agency did.
- **Journalists** want a named person, a named place, a document, and something that changed. Not a position paper.
- **Buyers and exporters** want risk to their supply chain, quantified.
- **Local politicians** want to know who in their district is affected and how many of them there are.
- **International bodies** want the state's obligation, the breach, and the evidence, set out briefly.
- **Your own group** needs to see that the fight is winnable and that this month's work mattered, or it will not be there next year.

STEP 5: MEDIA

Coverage raises the cost of a decision and reaches people no filing reaches.

Build the story

"GMOs are dangerous" is not a story; it is an opinion an editor has seen a thousand times. These are stories:

- The regulator approved a release using a document that got the local geography wrong, and here is the photograph.
- The monitoring that was promised as a condition of approval has never been done, and here is the freedom of information reply saying so.
- The government will not release the safety data it relied on.
- Farmers who will lose their certification were never notified.
- The consultation was held in a language most affected people do not read, with eleven days' notice.

Every one of those is specific, checkable, and about a document. Lead with your strongest verified fact and offer a named human voice — a farmer, a beekeeper, an elder, a local official — who will speak on the record.

Reach outlets in rings

Start local: community radio, local papers, farming press. The farming and trade press matters more than general news here, because the people who make the commercial decisions read it and the general public does not.

Then national. Then international, and then the specialist international channels: biosafety information services, organic and food-sovereignty networks, seed-saver networks, and the science press, which will cover a process failure carefully if you can document it.

Coverage in one ring pulls in the next. A local story about eleven days' consultation notice becomes a national story about how consultation is run, which becomes a question in parliament.

Sustain it, and keep it true

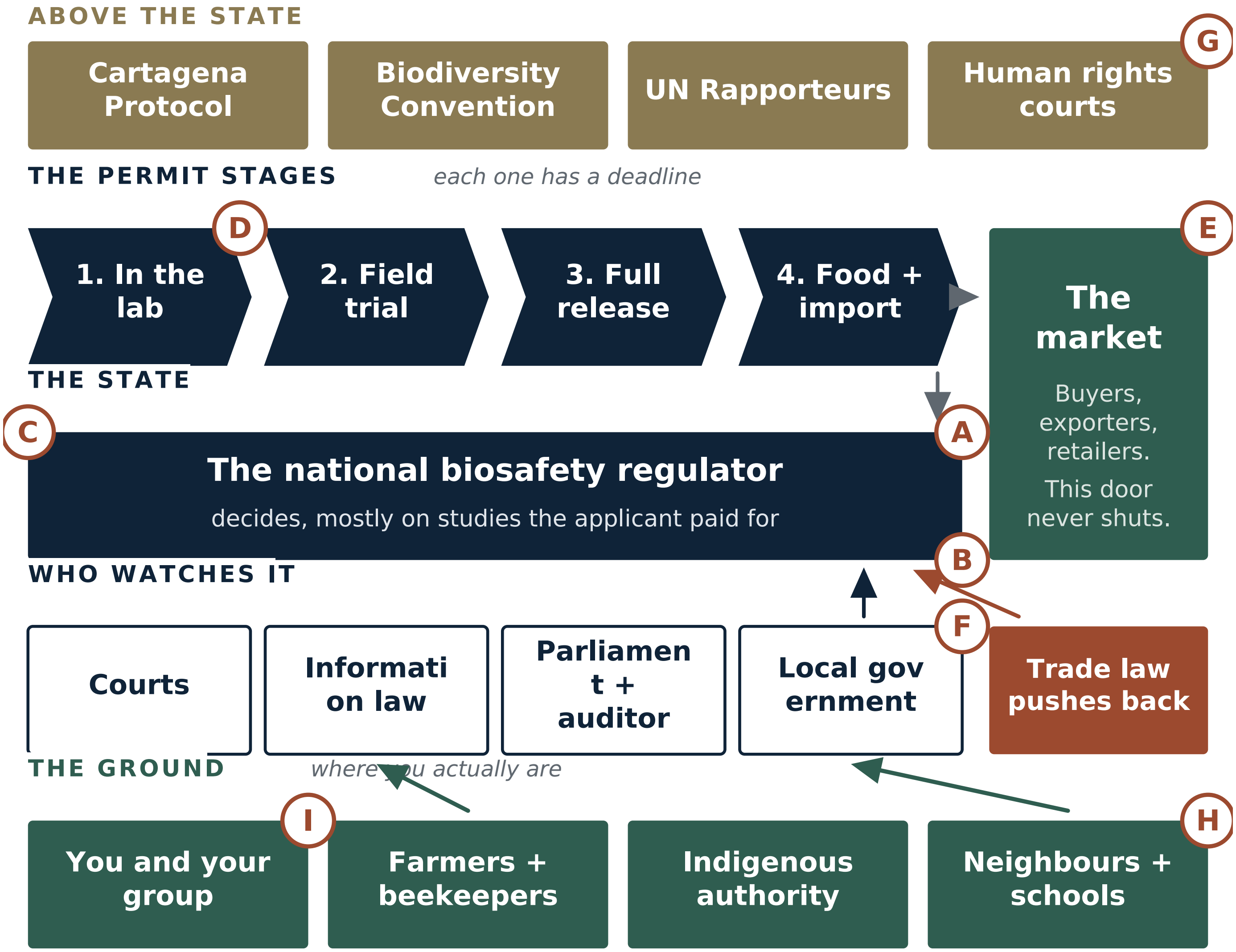
One article changes little. A sequence changes decisions. Plan your hooks in advance so there is always a next one: the application, the gap you found, the comment deadline, the number of submissions, the freedom of information reply, the council motion, the decision, the appeal.

And hold the line on accuracy without exception. One exaggerated claim — a health scare you cannot support, a number you did not check, a document you described from memory — will be found, and it will be used to dismiss the twenty things you got right. Being the accurate ones is not a moral pose here. It is the whole basis of your leverage, because your opponent has money and lawyers and you have a file that holds up.

LETTERS AND EMAILS YOU WILL NEED

Every one of these does two jobs: it asks for something, and it creates a record that you asked. Keep a copy of everything you send and everything you receive. Send anything important by a method that proves it arrived. Log every item in your master timeline.

Replace anything in square brackets. Keep them short. Attach your evidence rather than describing it.



NINE TEMPLATES, NINE PLACES THEY LAND

- (A) Public comment on the application
- (B) Freedom of information request
- (C) Ask to extend the window
- (D) Questions to the applicant
- (E) Warning to a buyer
- (F) Motion for a local council
- (G) Submission to an international body
- (H) Tip to a reporter
- (I) Public statement

Figure 6. Which letter goes where. Nine templates, nine positions on the same field. If you can see where a letter lands, you can see what it is supposed to do.

TEMPLATE A — Public comment on an application

To: [the regulator] **Subject:** Public comment on application [number] — [applicant], [organism and trait], [location]

I am writing about application [number], on which comments close on [date].

I am [who you are and how you are affected — “a certified organic maize farmer whose land lies 1.8 km from the proposed release site”].

I ask that the application be [refused / deferred pending independent assessment / granted only subject to the conditions below], for the following reasons.

1. [Specific gap.] The risk assessment states at page [N] that [claim]. This is not correct. [Evidence, attached as Annex 1.]

2. [Second gap.] The application contains no [monitoring plan / coexistence measures / liability provision / gene flow assessment for [species]]. [Why that matters here, in one or two sentences.]

3. [Process.] [Consultation was notified on [date], giving [N] days; affected farmers within [N] km were not individually notified.]

I request that the agency: (a) [extend the comment period / hold a public hearing in [place]]; (b) require an independent assessment of [the specific question]; and (c) place on the record its response to each point above.

Attached: [list of annexes].

[Name, address, date, signature]

TEMPLATE B — Freedom of information request

To: [the agency’s information officer] **Subject:** Request for information under [name of the law]

Under [the law], I request copies of the following records:

1. The monitoring reports submitted under condition [N] of [licence/permit number] for the years [range].
2. All correspondence between [the agency] and [the applicant] concerning [specific subject] between [date] and [date].
3. The minutes of the [committee] meeting held on [date] at which [decision] was considered.

If any part is withheld, please state the specific exemption relied on for each withheld document and provide the remainder.

I would prefer to receive these electronically. Please confirm receipt and the date by which you expect to reply.

[Name, contact details, date]

TEMPLATE C — Request to extend the window or hold a hearing

To: [the regulator] **Subject:** Request to extend the comment period on application [number] and to hold a public hearing

The comment period on application [number] opened on [date] and closes on [date], a period of [N] days. The application and its annexes run to [N] pages, and [N] were published only on [date].

The people most directly affected — [describe: farmers within the release area, the community at [place]] — were not individually notified, and the notice was published only in [language / outlet].

We request an extension of [N] days and a public hearing at [place], so that affected persons can be heard. We are [N] individuals and [N] organisations, listed below.

[Names]

TEMPLATE D — Letter to the applicant or company

To: [company / institute] **Subject:** Application [number] — questions on the record

We are [who]. We are participating in the consultation on your application [number].

We ask you to answer the following on the record:

1. Were any of the studies relied on in the risk assessment conducted independently of [company]? Please identify them.
2. What is your plan if [the trait] is detected outside the release area, and who bears the cost?
3. Will you indemnify neighbouring certified producers against loss of certification arising from your release? If not, why not?
4. [Specific question about the gap you found.]

We will publish your response, or the fact that there was none.

[Names, date]

TEMPLATE E — Letter to a buyer, retailer or exporter

> **To:** [buyer / retailer / export body] **Subject:** Supply risk from proposed [organism] release in [region]
>
> We are writing about a pending approval for [organism and trait] at [place], which is within the sourcing area for [product].
>
> [Market] does not accept this trait. [Cite the importing country's rule or the certification standard.] Once released, the trait cannot practicably be kept out of the regional supply, because [pollen range / shared handling and storage / seed movement].
>
> The comparable precedent is [StarLink maize in 2000 / LibertyLink rice in 2006], where contamination closed export markets and produced litigation lasting years.
>
> We are asking you to state publicly, before the decision on [date], that you will not source [product] carrying this trait, and to write to [the regulator] setting out the commercial risk.
>
> [Names, contact, date]

TEMPLATE F — Motion for a local council

> **That this [council] notes the pending application [number] for the release of [organism] at [location] within our [district];
>
> notes that [N] certified organic producers, [N] beekeepers and [N] households lie within [N] km of the proposed site;
>
> notes that affected residents were given [N] days' notice and were not individually notified;
>
> resolves to write to [the regulator] requesting [an extension of the comment period / a public hearing in this district / refusal of the application];
>
> and requests that officers report back on what powers this [council] holds in relation to [planning permission / use of council land / local notification requirements] for such releases.**

TEMPLATE G — Submission to an international body

> **To:** [UN Special Rapporteur on the right to food / on toxics / on the rights of indigenous peoples — or your Cartagena Protocol national focal point] **Subject:** [Country] — approval of [organism] release at [place] without [adequate assessment / consultation / consent]
>
> We are [who]. We bring to your attention the following.
>
> **The facts.** On [date], [agency] approved [decision] permitting [what], at [where], affecting [whom].
>
> **The concern.** [Two or three sentences: what was not done that should have been. Consultation, consent, independent assessment, socio-economic assessment under Article 26 of the Cartagena Protocol, monitoring, liability.]
>
> **What we have done domestically.** [Comments filed on [date]; information request [number] refused on [date]; appeal lodged [date].] Domestic remedies are [exhausted / ineffective] because [reason].
>
> **What we ask.** That you [seek information from the State / raise this with the Government / consider it in your forthcoming report].
>
> Evidence attached: [list].

TEMPLATE H — Email to a reporter

> **To:** [journalist] **Subject:** [Specific finding] — documents attached
>
> I follow your reporting on [beat].
>
> Short version: [one sentence with the specific, checkable finding. "The risk assessment for a release approved last month says there are no compatible wild relatives at the site. Here is a dated photograph of one, 300 metres from the fence."]
>
> I have the application, the risk assessment, and [the freedom of information reply / the photographs / the signed statements from N farmers]. I can put you in touch with [named farmer] and [named scientist], both of whom will speak on the record.
>
> The decision date is [date], so there is a peg.
>
> Can I send you the file?
>
> [Name, phone]

TEMPLATE I — Public statement

> **[Group name] on the proposed release of [organism] at [place]**
>
> On [date], [agency] will decide on application [number].
>
> We are [N] farmers, beekeepers, residents and organisations from [place]. We have read the application. It does not [state the single clearest gap].
>
> We are not asking for the impossible. We are asking for [the specific ask: an independent assessment, a public hearing, a monitoring requirement with funding attached, a liability rule].
>
> [Named person], who farms [N] hectares at [place], said: “[one plain sentence].”
>
> Contact: [name, phone, email]. The documents are available at [link].

IF YOU HAVE ONLY A FEW HOURS A WEEK

In order of what you get back for the time you put in:

1. **Find the file and the deadline.** Search the Biosafety Clearing-House and your national register. One evening. Everything else depends on it.
2. **Read the risk assessment and write down one thing it does not answer.** Two evenings. You are not looking for fraud, just an unanswered question.
3. **File a comment before the deadline.** One evening. Specific, signed, attached evidence, with a clear ask. This is the highest-value hour in the whole guide.
4. **Get three affected people to file their own, in their own words.** A week of phone calls. Multiplies the weight of point 3.
5. **Send one freedom of information request** for the monitoring reports or the independent studies. Thirty minutes. Works while you sleep.
6. **Email one reporter and one buyer.** An hour. Templates E and H.
7. **Ask a local councillor to put Template F on the agenda.** One meeting.

Do only the first three and you have done more than most objections ever manage: you have put a specific, answerable defect on the official record before the decision was made. Everything after that builds on it.

WHEN THE SYSTEM IS TILTED

It usually is, in specific and recognisable ways. Knowing which way it tilts tells you where not to waste your year.

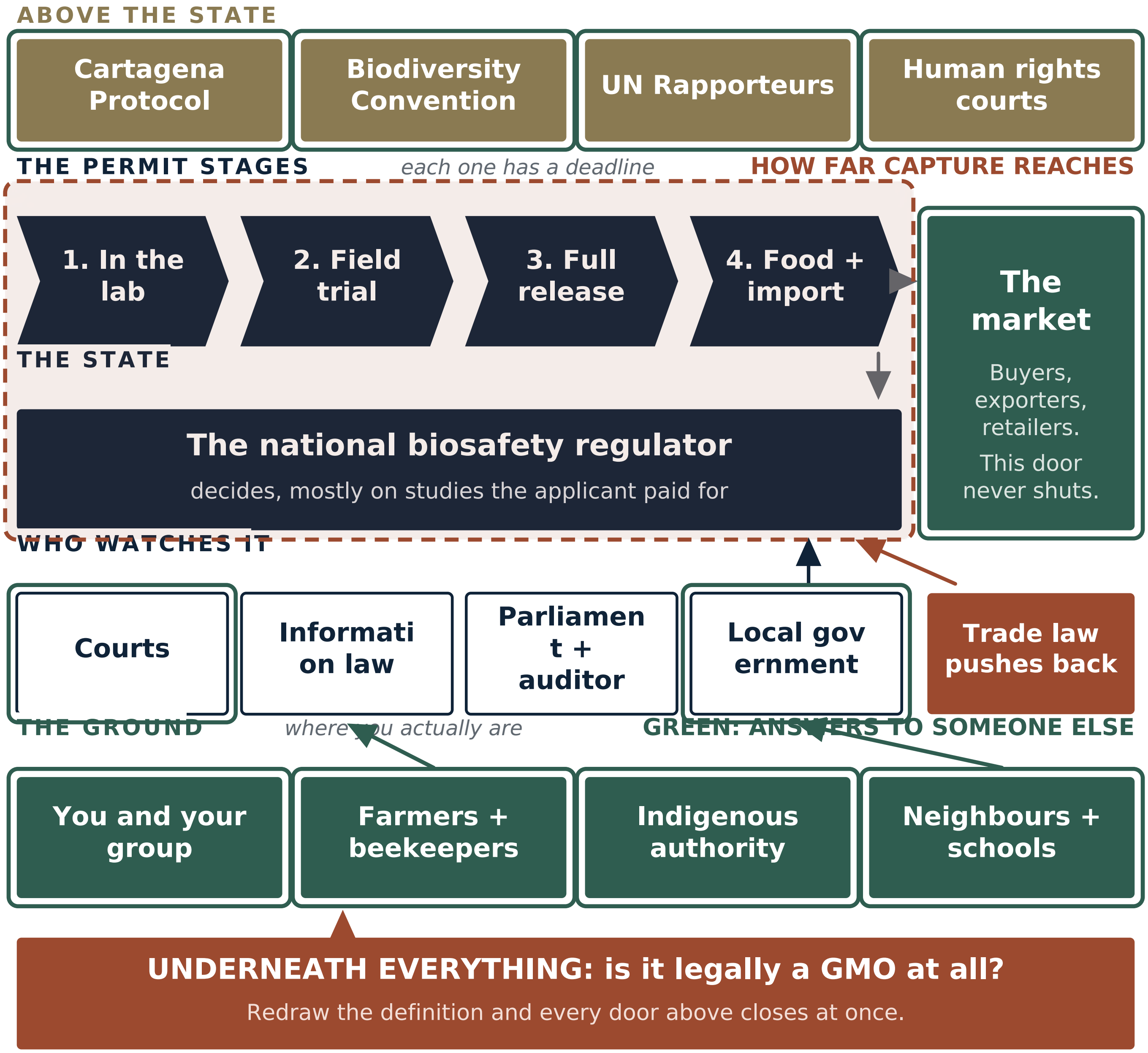


Figure 7. How far capture usually reaches. The shaded zone covers the permit stages and the regulator, where the applicant's money and the revolving door do their work. Everything outlined in green answers to someone else — voters, judges, customers, or other countries — which is exactly where a captured system is weakest.

How the tilt shows

- **The applicant supplies the data.** In most systems, the studies the regulator reviews were produced or paid for by the company seeking approval. Independent replication is rare and rarely funded.
- **The same people move back and forth** between the regulator, the industry and the research institutes that develop the products. Look at the committee membership and the previous employers. This is public and it is checkable.
- **Large parts of the file are blacked out** as confidential business information, including sometimes the very data on which safety is claimed.
- **The consultation is technically real and practically useless** — short notice, in one language, on a website nobody affected reads, with no obligation on the agency to change anything.
- **The scope is written narrow.** Australia's regulator, for example, is explicitly barred from considering trade, marketing, food safety or the benefits of a product when assessing a licence. Arguments outside the scope are simply not read. Find out what your regulator is permitted to consider before you write, and put the out-of-scope arguments somewhere they

will land — the parliament, the press, the market.

- **The category is redrawn.** The most effective way to end an argument about approving engineered organisms is to decide they are not engineered organisms. This is happening now, in several jurisdictions at once, and it is the single biggest structural change in this field.

What still works when it is tilted

- **The record.** An agency that must answer your specific point on the record is constrained by its own answer afterwards. Weak answers become the ground for appeals.
- **Procedure.** Captured agencies still have to follow their own rules, and they are often sloppy about it precisely because they expect no scrutiny.
- **The gap between promise and delivery.** Approvals come with conditions. Conditions are rarely monitored. That gap is documentable and it is where a surprising number of cases turn.
- **Levels the capture does not reach.** Local councils. Courts. Export buyers. Foreign regulators. Funders' own safeguard offices. International bodies. Insurance companies. Shareholders.
- **The market**, which does not care about the regulator's opinion at all.
- **Documenting the capture itself.** Committee members' affiliations, the redactions, the eleven-day consultation — these are not side issues. In front of a court, a parliament, or an international body, they are the evidence that the domestic process could not be relied on, which is exactly what those forums need to hear before they will act.

WHEN IT IS ACTUAL CORRUPTION

Sometimes the problem is not a tilted design but plain wrongdoing. The signatures are recognisable:

- An approval granted unusually fast, or against the written advice of the agency's own reviewers.
- Committee members with undisclosed financial interests in the applicant.
- Field trials proceeding before the permit was issued, or outside the permitted area, or unreported.
- Public research money and public land being used to develop a product whose commercial rights sit with a private company, on terms that were never published.
- Consultation records that describe meetings that did not happen, or list attendees who say they were not there.

If you find any of this, change route. Do not argue the science. Document the irregularity precisely — dates, names, documents, who signed what and when — and take it to the channels that sit outside the agency: the national auditor, the anti-corruption body, the ombudsman, the public accounts or environment committee of your parliament, the funder's own investigations office, and the press.

Two practical notes. First, be careful and be exact; an accusation you cannot document is worse than no accusation, and in some countries it is dangerous. Second, keep the corruption argument and the technical argument separate in your filings, so that if one is dismissed the other still stands.

HOW IT ALL FITS TOGETHER

The five steps are not a sequence. They run at once, each feeding the others. Your documentation is what makes the comment credible, what makes the news story publishable, and what makes a lawyer take the case. The group is what makes the comment carry weight. The press coverage is what makes the council motion pass. The council motion is what makes the national agency answer.

Weeks 0 to 4 — find and file

Identify the application, the stage, the decision-maker and the deadline. Read the risk assessment. Write down what is missing. Get five affected people. File comments — yours and theirs. Send one information request. This month decides most of what follows, and almost all of it is free.

Months 1 to 12 — build and widen

Map the ground. Photograph it. Collect signed statements. Find your two scientists and your lawyer. Approach buyers and exporters. Get the local council motion. Build the media sequence. Appeal any refused information request. If the decision goes against you, this is when you decide — with legal advice — whether there is a procedural ground worth taking further.

Do the time-critical things first. A comment deadline does not come back. A field trial that has been planted cannot be un-planted. A release that spreads cannot be recalled. Everything that can only be done before a date should be done before anything that can be done at any time.

Years 1 to 5 — the case and the second lap

If it went to court or to an international body, this is the slow part. Keep the group meeting. Keep the file current. Keep the press sequence running so the story does not die between hearings.

And watch for the two moves that follow a win: the **appeal**, and the **re-application** with slightly altered paperwork. Watch also for the third and quietest move — the **rule change** that removes the whole category from regulation, which turns your victory into a technicality. Comment on rule changes with the same seriousness you gave the permit. When your government opens a consultation on how gene-edited organisms should be classified, that consultation is worth more than ten permit fights.

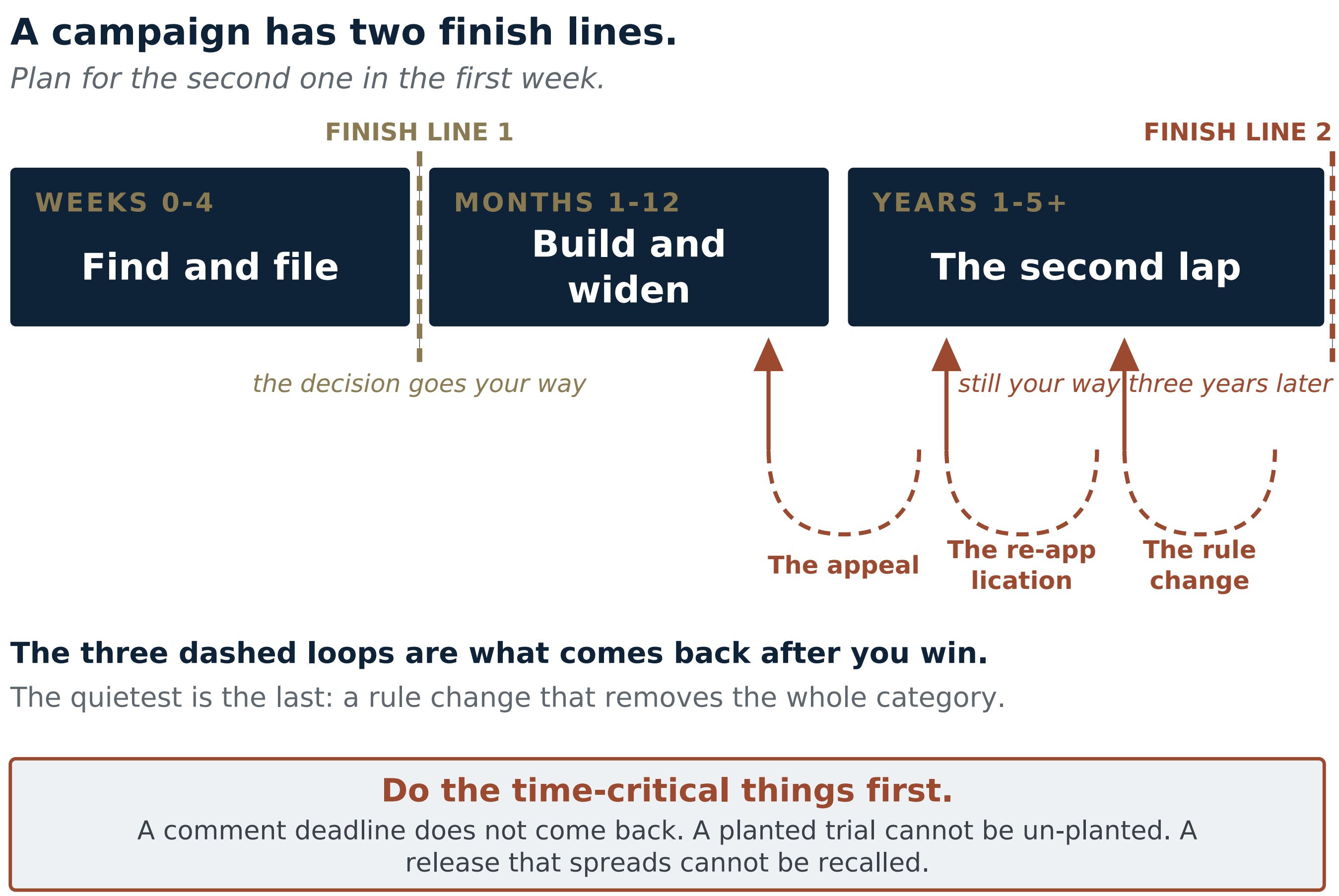


Figure 8. The shape of a campaign, and its two finish lines. Almost everyone plans for the first one. The three dashed loops are what comes back at you after you win, and the last of them is the quietest and the most damaging.

FINAL ASSESSMENT

What you actually have, stated plainly:

Almost every decision in this field is made on paper, in public, with a deadline. That is the fact this entire guide rests on. Applications, risk assessments, committee minutes, licences and their conditions are documents, and in most countries you can get them. Somebody has to read them, and almost nobody does.

You win on gaps and process, not on danger. The successful cases turned on missing monitoring, absent consultation, unverified claims, conflicts of interest, and the precautionary principle where certainty was lacking. That is what regulators must answer and what courts will review.

Consent is the strongest ground where it applies. Where a release affects indigenous territory, free, prior and informed consent is a recognised standard with binding judgments behind it in more than one regional human rights system. Nothing else in this field carries that weight.

The market has no procedure and no deadline. Buyers, exporters and retailers have ended engineered products that regulators were preparing to approve. That door is always open and it costs nothing but postage.

And the limits are real. The technology is not going to be uninvented. Engineered crop area is growing, not shrinking. The process is slow, the data is supplied by the applicant, the agencies are frequently captured, trade law punishes the boldest wins, and the definitions are being rewritten to shrink the whole field of regulation. Anyone who tells you otherwise is selling you something.

So the honest promise is this. Find the application. Read it. Name what is missing, with a page number. Get the people it affects to sign next to you. File before the deadline. Ask for the documents they did not publish. Tell one reporter and one buyer. Take the council motion. Keep the file, keep the group, and be there for the appeal.

Do that, and you will not end genetic engineering. You may well stop one release, in one place, and put on the permanent public record a defect that every future application has to answer. Farmers' networks with no money have done exactly this and won — in the Philippine courts, in the Indian consultations that produced a moratorium still standing after fifteen years, in the Mexican lawsuit that became a constitutional amendment, and in a Burkinabè village whose objection closed a programme funded from three continents.

None of them started with more than someone reading a document nobody else had read.

GLOSSARY

Advance informed agreement (AIA) — the Cartagena Protocol procedure requiring an exporter to notify, and the importing country to decide, before a living modified organism is first sent there for release into the environment.

Biosafety Clearing-House (BCH) — the free public database at bch.cbd.int holding national biosafety decisions, laws, risk assessments and the contact details of each country's deciding authority.

Bt — short for *Bacillus thuringiensis*, a soil bacterium. "Bt crops" are engineered to produce one of its insecticidal proteins.

Cartagena Protocol on Biosafety — the international agreement, in force since 2003 with 173 parties, governing the movement of living modified organisms between countries. The United States is not a party.

Centre of origin / centre of diversity — the region where a crop was domesticated and where its traditional varieties and wild relatives still exist. Releases in these regions are treated as higher risk.

Coexistence rules — the rules, where they exist, for keeping engineered and non-engineered production separate, and for who bears the cost.

Confined / field trial — an outdoor release under conditions meant to limit spread. Usually the earliest outdoor stage, and the easiest one to challenge.

Event — a specific engineered version of an organism, identified by a code such as MON 810. Approvals are granted event by event.

Free, prior and informed consent (FPIC) — the standard that indigenous peoples must give consent, before the fact and with full information, to projects affecting their lands. Recognised in the UN Declaration on the Rights of Indigenous Peoples and enforced in regional human rights courts.

Gene drive — an engineered system designed to spread a trait through a wild population faster than ordinary inheritance would. Self-spreading and, in practice, not recallable.

Gene editing / new genomic techniques (NGTs) — newer methods, such as CRISPR, that alter an organism's own DNA. Several countries have moved these outside their GMO rules.

GMO / LMO — genetically modified organism; living modified organism. "LMO" is the term the Cartagena Protocol uses.

Herbicide tolerance — engineering that lets a crop survive a herbicide that would otherwise kill it. The most widespread engineered trait by area.

Precautionary principle — the principle that a lack of full scientific certainty is not a reason to postpone measures against serious or irreversible harm. Written into many national laws and constitutions, and the basis of the 2024 Philippine ruling.

Risk assessment — the document the applicant supplies and the regulator reviews. Reading it, and finding what it does not address, is the core work of this guide.

Writ of kalikasan — a Philippine legal remedy protecting the constitutional right to a balanced and healthful ecology. The tool MASIPAG used.

YOUR FIRST WEEK: A CHECKLIST

- [ ] Search bch.cbd.int for your country and the organism.
- [ ] Find your national biosafety authority and its register of applications.
- [ ] Sign up for its email alerts, if it has them.
- [ ] Find the application number, the applicant, and the closing date.
- [ ] Write the closing date somewhere you will see it every day.
- [ ] Download the application and the risk assessment. Save them as PDFs.
- [ ] Read the risk assessment. Write down three questions it does not answer.
- [ ] Find out where exactly the site is. Go and look at it if you can.
- [ ] List every farm, apiary, house, school and water source within 5 km.
- [ ] Find two people who will do this with you.
- [ ] Send one freedom of information request.
- [ ] Draft your comment using Template A.
- [ ] Ask three affected people to write their own, in their own words.
- [ ] File before the deadline.
